BUILDGHOST

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Architectural Decision Review

Architectural Decision
Why I went with Electron over Tauri for my GhostOps Terminal Build.

When I was laying the groundwork for GHOSTops, I looked hard at the desktop framework landscape. Everyone is drooling over modern, lightweight stuff like Tauri and Wails right now. And I get the hype. But while the whole industry is obsessing over tiny file sizes, I intentionally went with Electron.

Here is the raw engineering logic behind why.

The Shiny New Toys Tauri and Wails are honestly incredible. Instead of dragging an entire Chromium browser into the app, they piggyback on your OS's native web engine. The result? The apps are lightning fast and stupidly small. We are talking under 10MB compared to Electron's massive 150MB footprint. But here is the massive catch. They pull that off by being locked down in a heavy security sandbox. If I was just building a cute little chat app or a to do list, Tauri would be a no brainer.

The Reality of QA Engineering GHOSTops is not some basic desktop widget. It is a full blown QA automation and diagnostics suite. Sure, the early modules are doing surgical DOM targeting, but the future roadmap is heavy. We are talking API REST tools, automated test generation, and deep local file system stuff. Native webviews in Tauri and Wails actively fight against you when you try to do this. Trying to hack a Mac's WebKit to forcefully bypass strict security protocols or inject scripts into a foreign DOM is a miserable experience. The security sandbox is doing its job, which is keeping you out. And as a QA engineer, being kept out is the exact opposite of what I need.

Why Electron Actually Wins Yes, Electron bundles its own instance of Chromium and Node.js. Yes, it is a heavy beast. But it gives me absolute, god tier control over the browser environment and the local OS. I can spin up isolated browser windows and literally dictate the rules. I can nuke web security flags to allow cross origin testing and inject whatever custom scripts I want without the OS throwing a fit.
Plus, the deep Node.js integration means I can easily bridge the gap between a live web session and my local backend. That is mandatory for saving session logs, spitting out test files, and hooking up the local AI orchestration later on. There is a very good reason heavy hitters like Cypress are built on Electron. Having absolute control over Chromium is not a luxury in this space. It is a hard requirement.
The Verdict So yeah, I am knowingly trading file size and memory overhead for absolute, raw access. For the long term vision of GHOSTops, having that Chromium level control is the only way to build this thing right.